



A reduced toluene reference fuel chemical kinetic mechanism for combustion and polycyclic-aromatic hydrocarbon predictions



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ABSTRACT

A reduced toluene reference fuel (TRF, *n*-heptane, *iso*-octane and toluene)-polycyclic-aromatic hydrocarbon (PAH) chemistry mechanism with 109 species and 543 reactions is proposed for combustion, PAH and soot formation predictions. The mechanism has been validated with shock tube ignition delays, laminar flame speeds, species profiles in premixed flames, and with homogeneous charge compression ignition (HCCI) and direct injection compression ignition (DICI) engine combustion data. Sensitivity and reaction pathway analyses were used to identify the important reactions and reaction pathways for PAH formation for both non-aromatic and aromatic fuels. The results show that the mechanism yields good agreement under all validation conditions. The PAH species concentrations are well predicted for eight flame configurations (C_2H_4 , *n*-heptane, *iso*-octane, benzene ($C_6H_6-A_1$) and toluene). For the non-aromatic fuels, reactions related to C_2H_4 , C_2H_3 and C_2H_2 show the highest sensitivity for PAH formation, while with the aromatic fuel, PAH formation depends highly on reactions related to the ring structure. Benzene (A_1) is mainly produced through the combination of $C_4 + C_2$ and two C_3 radicals for non-aromatic fuels, and for aromatic fuels A_1 mostly comes from the fuel molecule (toluene). The formation pathways of higher PAH species are quite similar for both non-aromatic and aromatic fuels, including $C_5H_5 + C_5H_5 = A_2 + 2H$, the hydrogen abstraction acetylene addition (HACA) scheme, addition reactions between radicals and molecules and reactions between aromatic radicals and molecules. The mechanism is also capable of capturing combustion and emissions in HCCI and DICI engine simulations. The toluene content greatly enhances the PAH and soot formation with TRF fuels. However, improving mixing with TRF fuels due to their longer ignition delay also favors soot oxidation, and the final soot emissions of TRF fuels can be lower than diesel. The good agreement in the predictions of PAH species and soot emissions with the various fuels suggests that the current mechanism can be used reliably for combustion and PAH predictions for surrogate fuels of practical interest.

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1. Introduction

Fuel properties, including both physical and chemical properties, are very important parameters that affect the performance and emissions of internal combustion (IC) engines [1]. Due to more-and-more stringent pollutant emission and carbon dioxide (CO_2) legislation, highly premixed combustion concepts have become competitive measures to achieve clean and high efficiency combustion, such as HCCI [2] and reactivity controlled compression ignition (RCCI) [3], which also highlights the importance of

the fuel chemical properties, since the effects of the fuel physical properties are reduced due to the much longer available mixing time. Thus reliable chemical kinetic mechanisms are crucial for advanced engine combustion concepts simulations.

Currently diesel and gasoline are still the most widely used fuels in IC engines. Both are complex compounds of many components with different molecular structures, depending on the origin and refining process of the crude oil. In order to simulate the combustion processes of diesel and gasoline fuels, it would be ideal to apply a detailed kinetic mechanism, which contains all the possible components in the real fuels. However, currently it is still impossible to develop and apply such large mechanisms in computational fluid dynamic (CFD) simulations due to restrictions of computation power. Also, it is desirable to acquire insight into in-cylinder

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information from simulations to guide the development of combustion technologies. Thus, simplified surrogate mechanisms have been proposed to represent diesel and gasoline fuels by including the most important components in the fuels. For example, *n*-heptane and *iso*-octane, have been used as single surrogates for diesel and gasoline due to their similar ignitibilities [4]. But the drawbacks are also obvious since no other components are considered in these simplified approaches.

Many researchers have also pointed out that aromatics play important roles in soot formation processes, and the aromatic content in diesel and gasoline fuels is usually around 25–35% [5]. Thus, more realistic multi-component surrogate models, which take the aromatic components into consideration, have been suggested and proposed for both diesel and gasoline fuels. For example, a four component surrogate model including *n*-decane, *iso*-octane, methylcyclohexane and toluene has been suggested by Farrell et al. [6] as a diesel surrogate, while for gasoline fuel, a three component TRF surrogate model, including *n*-heptane, *iso*-octane and toluene, is a widely used surrogate [7,8]. Toluene is present in both diesel and gasoline surrogate models, which confirms the importance of the toluene component for the development of reliable surrogate models. Since the ignition delays of paraffins from C₇ to C₁₆ are quite similar [5], it is possible to replace the *n*-decane component in diesel fuel surrogates with *n*-heptane, a simpler paraffin to reduce the complexity of the surrogate model. Therefore, the TRF model (*n*-heptane/*iso*-octane/toluene) can be regarded as a foundation for the development of more realistic and reliable surrogate models. In addition, since PAH species are considered as soot formation precursors in many soot modeling approaches [9], and the ring structure in the toluene molecule greatly affects the formation of PAH, it is also desirable to include toluene for better PAH and soot predictions.

Experimental and kinetic studies have been conducted to study the oxidation processes of primary reference fuels (PRF, mixture of *n*-heptane and *iso*-octane) and detailed and reduced PRF mechanisms are available from the literature [4,8,10–19]. The general oxidation reaction pathways proposed by Curran et al. [10,11] have been adopted in many kinetic studies. Currently mechanism validation has been shifting from solely predicting ignition delays to studying laminar flame speeds, and species predictions in various combustion devices to improve the prediction accuracy and application range of the mechanisms. The authors have also developed a reduced PRF mechanism [20] based on the detailed gasoline surrogate mechanism developed by Mehl et al. [8], which has been extensively validated with various experimental data. This reduced PRF mechanism is also used as the base mechanism for the development of the current TRF-PAH mechanism.

Although it has been recognized that toluene is an important component for diesel and gasoline surrogate models, the availability of toluene-related experiments and mechanisms is not as rich as for the PRF mechanism. Reduced toluene mechanisms have been validated against ignition delays under high temperature conditions and little attention has been paid to the predictions of laminar flame speed and species profiles in flames, which are important when a toluene sub-mechanism is applied for soot prediction. Therefore, in recent years many kinetic modeling and experimental studies related to toluene and TRF fuels, including ignition delays [21–23], laminar flame speeds [24,25], flames [16–18,26] and HCCI [27] combustion, have been conducted, and many detailed kinetic mechanisms have been proposed based on these studies. Andrae et al. [27–29] proposed several TRF mechanisms. The latest one consisted of 137 species and 635 reactions and was based on their previous studies and the works of Sakai et al. [15]. It has been validated with ignition delays, flame speeds and also species profiles in flow reactors and jet stirred reactors (JSR). Zhong et al. [19] proposed a chemical kinetic mechanism consisting of 89

species and 355 reactions to predict the ignition and oxidation behavior of a five component gasoline surrogate model, including *iso*-octane, *n*-heptane, ethanol, toluene and *di*-*iso*-butylene. Although the ignition behavior in shock tubes was well predicted by the mechanism, large discrepancies could be seen in HCCI combustion simulation, i.e., much higher combustion rates (pressure rise rate) compared to the experiments. Wang et al. [30] proposed an *n*-heptane/toluene/PAH mechanism based on the reduced PRF and toluene mechanisms developed by Ra et al. [4,31]. The mechanism has been validated with ignition delays, important species related to PAH and soot formation in *n*-heptane and toluene flames, and also against HCCI and DICI engine combustion and soot emissions data. However, flame speed validation was not considered at that time, and an *iso*-octane sub-mechanism was also not included in the mechanism. The detailed gasoline surrogate mechanism developed by Mehl et al. [8] for *n*-heptane, *iso*-octane, toluene and C₅–C₆ olefins contains more than 1300 species and has been validated with experimental data from shock tubes, stirred reactors and rapid compression machines (RCM). However, such a large mechanism is even very difficult to apply for zero dimensional (0-D) simulations. Thus, mechanism reduction is inevitable before it can be applied in engine combustion simulations.

The inclusion of toluene not only accounts for important fuel components, but also for PAH and soot formation. Thus, it is also very important to study the effects of toluene/benzene on the PAH formation process. The detailed PAH mechanism developed by Slavinskaya et al. [32,33] was reduced and incorporated in the *n*-heptane/toluene mechanism to study the effects of toluene addition on the PAH and soot formation processes under direct injection spray conditions by Wang et al. [30,34]. It was found that the addition of toluene greatly promotes the formation of PAH and soot, but the longer ignition delay of toluene also extends the available mixing time and thus reduces the locally rich regions and enhances soot oxidation. Raj et al. [7] developed a detailed TRF mechanism for gasoline surrogate fuels and PAH prediction, which contains around 230 species and 1350 reactions. This mechanism has been validated for premixed laminar flames of *n*-heptane, *iso*-octane, benzene and ethylene and also for counter-flow diffusion flames of *iso*-octane/toluene and *n*-heptane/toluene mixtures. The major PAH formation pathways were also identified and discussed in their study. However, their mechanism is still too large to be applied in engine CFD simulations and no ignition delay and flame speed validations were presented in their study.

In the present study a reduced TRF-PAH mechanism is proposed for application in engine combustion, PAH and soot emission predictions. The major PAH formation pathways for different fuels (non-aromatic and aromatic fuels) are analyzed and discussed. The paper starts with the assessment of a previous *n*-heptane/toluene mechanism, followed by the formulation and validation of the current TRF-PAH mechanism, including ignition delays, laminar flame speeds, species profiles in flames and HCCI and DICI engine combustion and emission data. Sensitivity and reaction pathway analyses with respect to PAH formation for the various fuels are also extensively discussed.

2. Mechanism development

2.1. Assessment of a previous *n*-heptane/toluene mechanism

The reduced *n*-heptane/toluene/PAH mechanism proposed by Wang et al. [30] showed good agreement in predictions of ignition delays and PAH species profiles, and HCCI combustion of *n*-heptane/toluene mixtures was also well predicted. Figure 1 shows comparisons of *n*-heptane and toluene laminar flame speeds between experiments and predictions with the mechanism (the

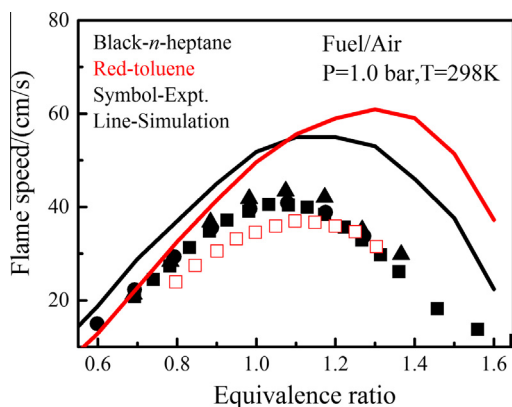


Fig. 1. Comparison of laminar flame speeds between experiments [24,35] and simulations for *n*-heptane and toluene with the reduced *n*-heptane/toluene mechanism of Wang et al. [30].

experimental data are taken from [24,35]). The simulations were conducted with the CHEMKIN Pro package [36]. The uncertainty for the *n*-heptane flame speeds from the experiments was around ± 1 cm/s, and the uncertainty for toluene was typically 1–2 cm/s. It is seen that although the ignition delay was well predicted by this mechanism, predictions of laminar flame speeds for both *n*-heptane and toluene are much higher (15–20 cm/s higher) than the experimental measurements.

2.2. Formulation of new TRF-PAH mechanism

The present TRF-PAH mechanism consists of three parts: (a) a base reduced PRF mechanism, (b) a reduced toluene sub-mechanism derived from the detailed LLNL gasoline surrogate mechanism [8], and (c) a reduced PAH sub-mechanism derived from the detailed PAH mechanism developed by Slavinskaya et al. [33].

2.2.1. Base PRF mechanism

The reduced PRF mechanism developed by Wang et al. [20,37] was taken as the starting base mechanism, wherein the C_3 core mechanism was adopted from the AramcoMech 1.3 mechanism [38]. This updated core mechanism has been extensively validated and shows reliable performance under various validation conditions. It is used to serve as the common “base block” for all the sub-mechanisms involved in the TRF-PAH mechanism. The reduced PRF mechanism was built based on this core mechanism by constructing simplified reaction pathways from *n*-heptane and *iso*-octane to C_3 species, and the lower C number reactions were taken care of with the core mechanism. These validation results are also summarized in the following sections and details about the base PRF mechanism can be found in [20,37].

2.2.2. Toluene sub-mechanism

The detailed LLNL gasoline surrogate mechanism [8] was taken as the base mechanism for the toluene mechanism reduction. Direct Relation Graph with Error Propagation (DRGEP) [39,40], reaction pathway, rate of production and sensitivity analyses were applied for the mechanism reduction. The mechanism reduction starts with the DRGEP method to remove unimportant species and reactions from the base mechanism to obtain an intermediate mechanism; then reaction pathway, rate of production and sensitivity analyses were applied to identify the most important species and reactions. If possible, isomer lumping was used to further reduce the species number. After each step, the reduced mechanism was validated against either detailed mechanisms or experimental measurements. Reaction constant optimizations

were applied to improve the ignition delay predictions under some conditions. These steps were conducted iteratively until a reduced mechanism with the desired size and prediction accuracy was obtained.

During the reduction the laminar flame speed was chosen to be the major target since ignition delay prediction can be easily adjusted through reaction rate constant optimization, while prediction of flame speeds depends on both the reaction pathway and the core mechanism. Figure 2 shows predictions of laminar flame speeds of toluene with both the detailed and reduced standalone toluene mechanism (45 species and 205 reactions) (the experimental data are taken from [24,25]). The detailed mechanism tends to slightly over predict the laminar flame speed and the reduced 45 species toluene mechanism still provides reasonable predictions.

Figure 3 shows the major reaction pathways for the reduced toluene mechanism. The numbers in the boxes are the fractions of toluene consumption through the various pathways under different conditions. It is seen that the H atom abstraction from the methyl branch (from $C_6H_5CH_3$ to $C_6H_5CH_2$) is the most important reaction pathway for toluene consumption, followed by the $C_6H_5CH_3 + OH = C_6H_5OH + CH_3$ pathway. Although not shown in this generic reaction pathway diagram, the reactions involving $C_6H_5CH_2 + O_2 \rightarrow C_6H_5CH_2OO$ and then the following products, such as $C_6H_5CH_2O$, C_6H_5CHO and C_6H_5O , are also included in the reduced toluene mechanism. As can be seen in Fig. 3, C_6H_6 (A_1) and C_6H_5 (A_1) are very important intermediate species in the toluene oxidation process, which inevitably affects the PAH formation process, since for fuels without ring structures, the pathway for the first ring formation can be quite different from that of toluene.

2.2.3. PAH sub-mechanism

The reduced PAH mechanism developed by Wang et al. [34] was applied for PAH formation predictions. The parent mechanism developed by Slavinskaya et al. [33] considered many possible PAH formation pathways, including the hydrogen abstraction acetylene addition (HACA) scheme [41], hydrogen atom migration, free radical addition schemes, methyl substitution pathways and reactions between aromatic radicals and stable aromatic molecules. This mechanism has been reduced to describe the PAH formation process up to pyrene (A_4). The building block formation reactions, such as the reactions involving C_4H_2 , *n*- C_4H_3 , C_4H_4 , *i*- C_4H_5 and C_5H_5 , were mostly retained in the reduced mechanism. The major formation pathways for the first ring are the recombination of $C_4 + C_2$ radicals ($C_2H_3 + C_4H_4 = A_1 + H$, $C_2H_2 + i-C_4H_5 = A_1 + H$ and $C_2H_3 + n-C_4H_3 = A_1$) and two C_3 radicals ($2C_3H_3 = A_1$). The formation of

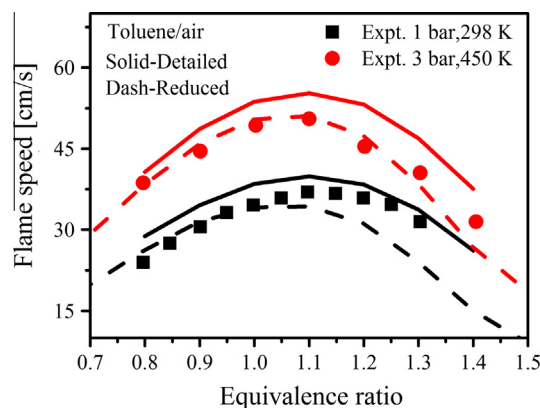


Fig. 2. Laminar flame speeds predicted by the LLNL gasoline surrogate mechanism and the standalone reduced toluene mechanism. Experimental data are taken from [24,25].

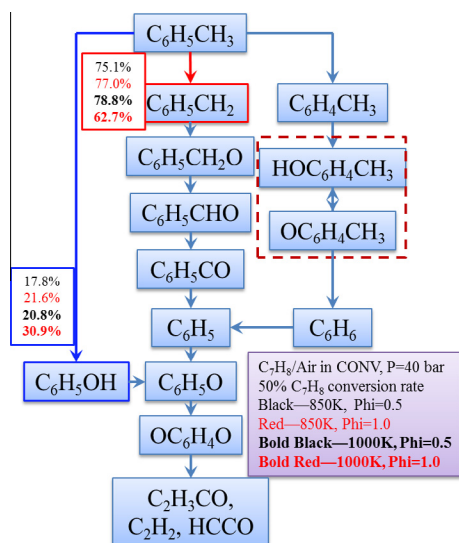


Fig. 3. Schematic reaction pathways for the reduced toluene mechanism.

mono-substituted aromatics, such as A_1C_2H and $A_1C_2H_3$, were also considered. Naphthalene (A_2) was mainly formed through the $2C_5H_5 = A_2$ and $i-C_4H_5 + A_1 = A_2 + 2H + H$ reactions, and the formation of larger PAHs, including phenanthrene (A_3) and A_4 , are described through the HACA sequence, small radical and molecule addition sequences and reactions between aromatic radicals and molecules. Detailed information and analysis related to the PAH formation pathway for various fuels are discussed in the following sections.

2.2.4. Formulation of TRF-PAH mechanism

The reduced TRF-PAH mechanism is formulated by combining the above-mentioned sub-mechanisms, resulting in an integrated mechanism with 109 species and 543 reactions. Duplicated reactions in both the toluene and PAH sub-mechanisms were removed from the integrated mechanism, and the simplified toluene oxidation sub-mechanism only considers the pathways from the toluene molecule to the C_3 and C_2 species. Since A_1 and related species and reactions exist in both the toluene and PAH sub-mechanisms, duplicated reactions in the PAH sub-mechanism (around 10 reactions) were removed from the integrated mechanism. In addition, a reduced NOX mechanism with 4 species and 12 reactions was also incorporated into the mechanism [42]. The reduced mechanism retains the thermal mechanism pathways and the N_2O -intermediate mechanism pathways. The PAH sub-mechanism in the current mechanism contains PAH formation and oxidation pathways up to A_4 . The thermodynamic and transport properties for the PRF and toluene mechanisms were taken from the detailed LLNL gasoline surrogate mechanism [8], and for the PAH related species that were not presented in the LLNL mechanism, the properties were taken from Slavinskaya et al.'s detailed PAH mechanism [33]. The complete mechanism including the thermodynamic and transport properties in CHEMKIN format is provided as [Supplemental material](#).

3. Mechanism validation

The proposed mechanism has been validated using various experimental data, including shock tube ignition delays, laminar flame speeds, species concentration profiles in flames and with engine HCCI combustion data. In addition, combustion and emissions data from diesel, *n*-heptane and *n*-heptane/toluene DICI experiments are also used to validate the proposed mechanism.

3.1. Ignition delay

Validations of the PRF mechanism have been extensively discussed in the authors' previous works [20,37], thus only brief results and discussions are presented here. Figure 4 shows the validation for PRF ignition delay predictions, including neat *n*-heptane, *iso*-octane and mixtures of *n*-heptane and *iso*-octane. The experimental data are taken from [43–46]. Ignition delay is defined as the time of a 400 K temperature increase compared to the initial temperature. The simulations were performed with the CHEMKIN Pro package [36]. It is seen that very good agreement is obtained with the base PRF mechanism.

Figure 5 shows ignition delay validations for the toluene sub-mechanism. The experimental data measured by Herzler et al. [22] and Shen et al. [23] were taken to validate the mechanism. It is worth pointing out that currently there is no available ignition delay data for neat toluene under low temperature conditions. It is seen that the overall agreement can be regarded as acceptable, though the mechanism tends to slightly under predict the ignition delays of the *n*-heptane/toluene mixture. In addition to this two component mixture, ignition delays measured by Gauthier et al. [47] for two three-component gasoline surrogates (surrogate A: 56% *iso*-octane, 17% *n*-heptane and 28% toluene; surrogate B: 63% *iso*-octane, 17% *n*-heptane and 20% toluene) are also compared with the predictions of the current mechanism, and the results are shown in Fig. 6. Similar to Fig. 5, the trends are captured by the mechanism, but the predictions are still slightly shorter than the experimental data. This indicates that further improvements to the toluene sub-mechanism may still be needed when ignition delay measurements for toluene under low temperature conditions become available.

3.2. Laminar flame speeds

Figure 7 shows predictions of laminar flame speeds using the current mechanism compared to the experimental results of van Lipzig et al. [35], Davis et al. [24] and Johnston et al. [25] for *n*-heptane, *iso*-octane and toluene. Compared to the results predicted with the previous *n*-heptane/toluene mechanism (Fig. 1), a great improvement can be seen with the current mechanism. It is seen that the agreement between predictions and experiments is mostly satisfactory for all the tested fuels.

3.3. Species profiles in flames

In addition to ignition delays and laminar flame speeds, special attention has been paid to predictions of PAH species profiles for all three fuel components in the mechanism. The species profiles, especially species related to PAH and soot formation from eight different flames are taken to validate the proposed mechanism. Table 1 shows the flame simulation conditions in the current study. All flames were simulated with the CHEMKIN Pro software package [36]. The references for the flames are also listed in Table 1. These flames include a small molecule C_2H_4 , where the PAH formation depends on the core C_3 mechanism and PAH building block reactions and also on the PAH formation mechanism itself. Also, *n*-heptane and *iso*-octane, where the PAH formation pathways depend on decomposition reaction pathways, and finally benzene (C_6H_6) and toluene, where ring structures in the fuel molecule greatly affect the PAH formation rates and pathways.

3.3.1. C_2H_4 flame

Figure 8 shows predictions of important intermediate species and PAH species profiles in the C_2H_4 flame with the current mechanism. The x-axis is the height above the burner (HAB) position. It is seen that good agreement can be obtained with the proposed

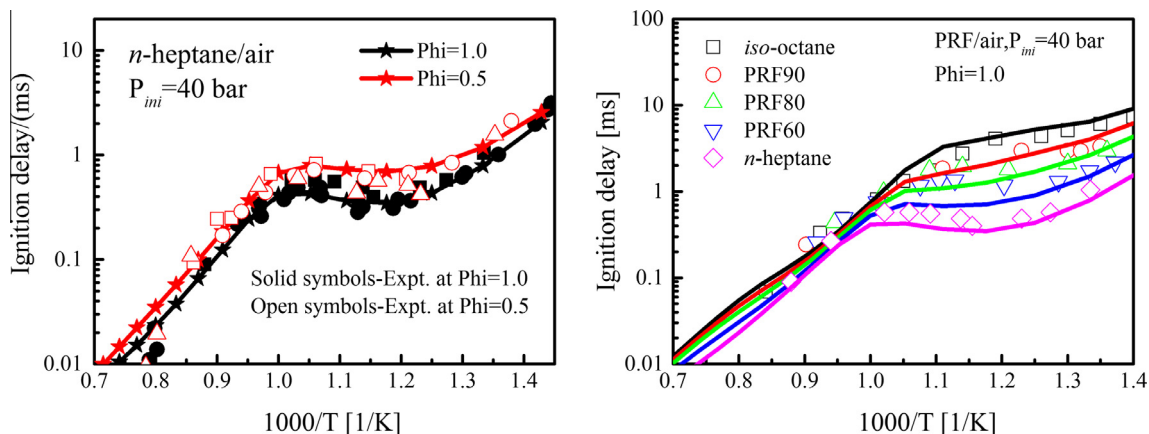


Fig. 4. Validations of the base PRF mechanism for ignition delay predictions. Experimental data are taken from [43–46].

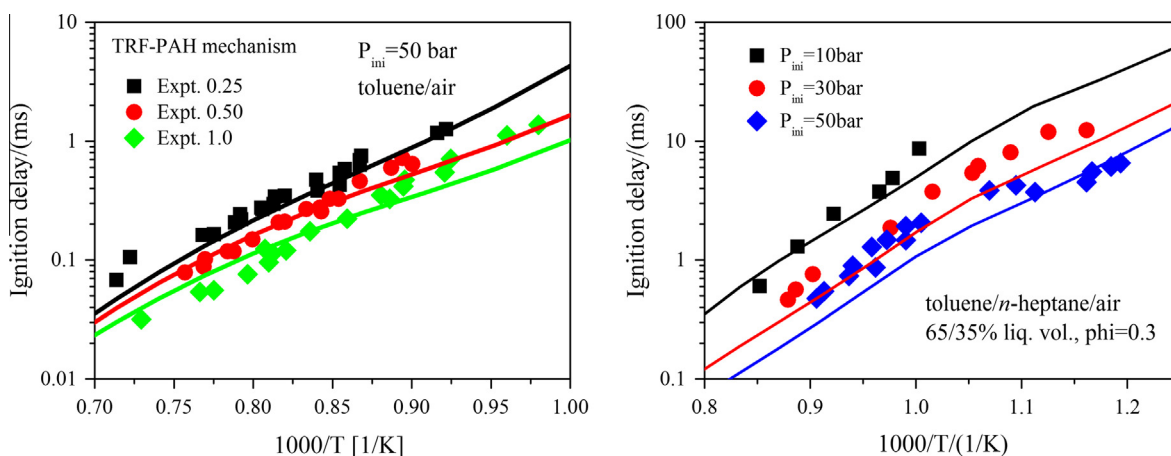


Fig. 5. Validations of the reduced toluene mechanism for ignition delay predictions. Experimental data are taken from [22,23].

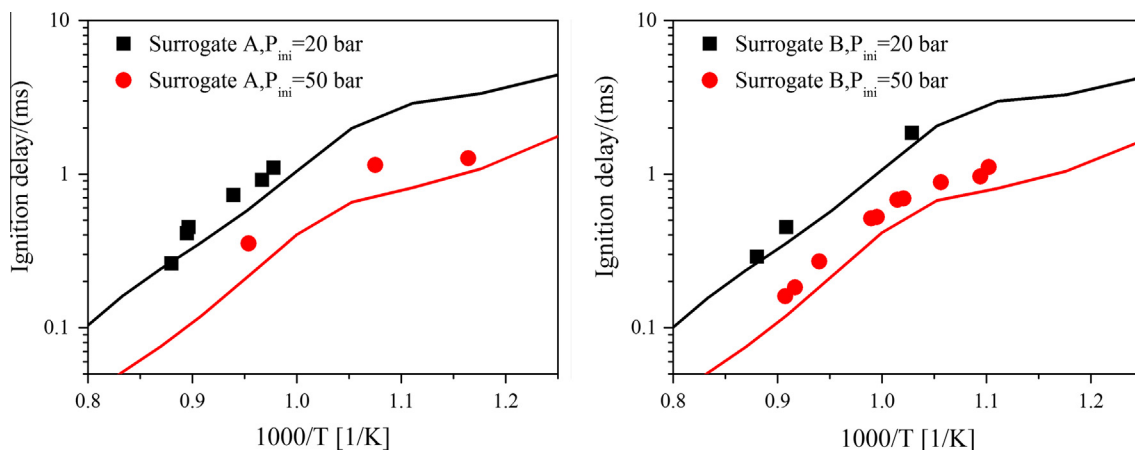


Fig. 6. Ignition delay validations for two gasoline surrogates with the proposed TRF-PAH mechanism. Experimental data are taken from [47].

mechanism. As mentioned above, the formation of PAH species in C_2H_4 depends highly on the core and PAH building block reactions, and the results shown here prove the reliability of the current core mechanism.

3.3.2. *n*-heptane flame

Comparisons of important intermediate species and PAH species between experiments and simulations in *n*-heptane flames (Flame 2 in Table 1) are shown in Fig. 9. C_2H_2 is considered as

one of the most important species for PAH and soot formation and prediction, and was chosen as one of the key species in the mechanism reduction processes. It is seen that the predictions of the current mechanism agree quite well with the experimental results for both intermediate and PAH species.

3.3.3. *iso*-octane flame

Figure 10 shows the species profiles predicted by the current mechanism in *iso*-octane flames as compared with experimental

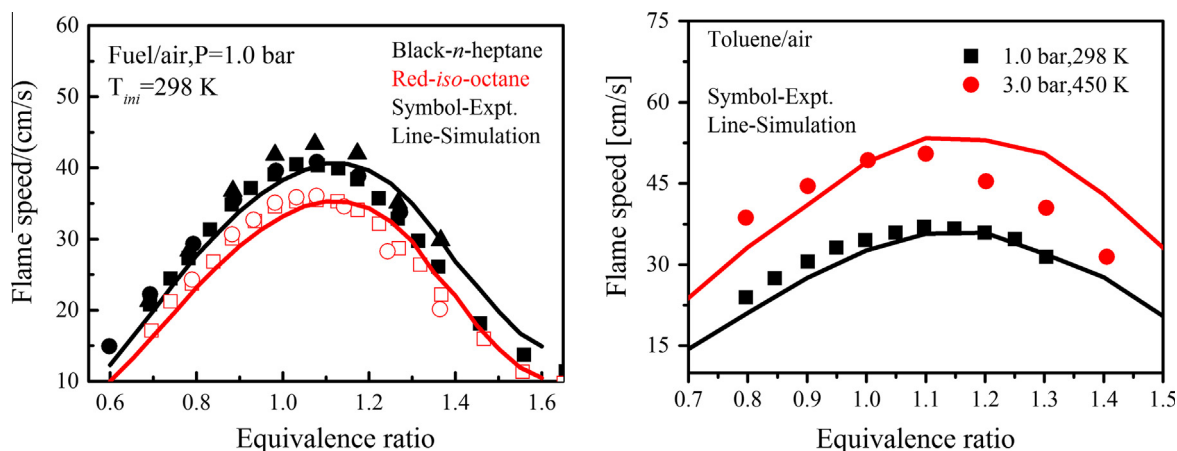


Fig. 7. Validations of the proposed mechanism for laminar flame speed predictions. Experimental data are taken from [24,25,35].

Table 1

Flame conditions simulated in the current study.

Flame No.	Fuel	Composition [mole frac.]			g/(cm ² s) or cm/s	P	Φ	Ref.
		Fuel	O ₂	N ₂ /Ar				
1	C ₂ H ₄	0.213	0.209	0.578/Ar	0.0084	1 bar	3.06	[48]
2-a	n-C ₇ H ₁₆	0.0398	0.2301	0.7301/N ₂	4.98	1 bar	1.9	[16,49]
2-b	n-C ₇ H ₁₆	0.0550	0.2879	0.6571/Ar	5.17	1 bar	–	[50]
3	i-C ₈ H ₁₈	0.0423	0.278	0.6797/N ₂	4.18	1 bar	1.9	[16,49]
4-a	Benzene	0.0957	0.4043	0.50/Ar	0.0021	4.0 kPa	1.78	[51]
4-b	Benzene	0.118	0.442	0.44/Ar	35.0	5.0 kPa	2.0	[52]
5	Toluene	0.05	0.45	0.50/Ar	35.0	4.0 kPa	1.0	[17]
6	CH ₄ toluene	CH ₄ :0.049 C ₇ H ₈ :0.015	0.234	0.702/N ₂	34.2	5.3 kPa	1.0	[53]

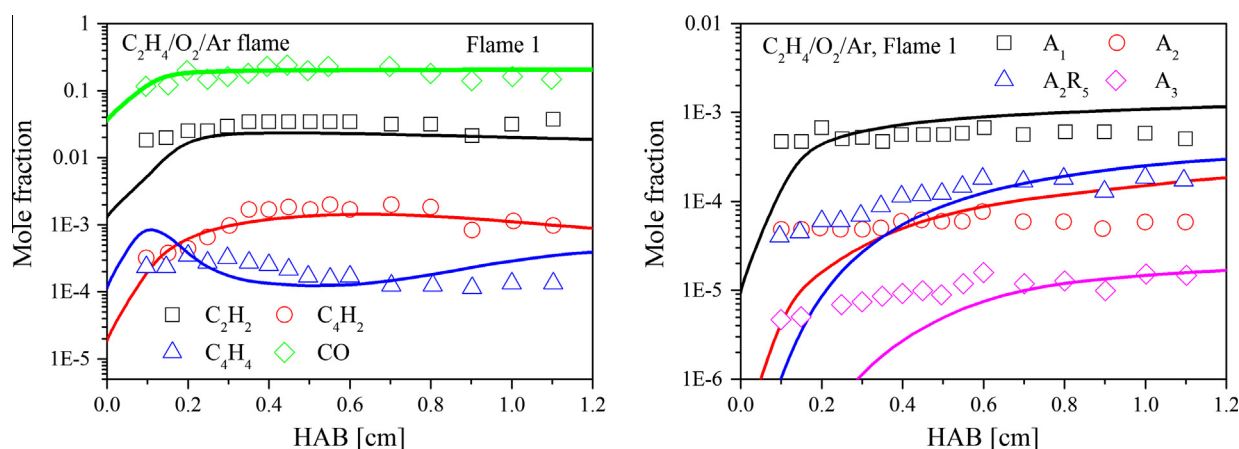


Fig. 8. Predictions of species profiles with the current mechanism in C₂H₄ flame. Experimental data are taken from [48] (Flame 1 in Table 1).

results. Again the agreements between simulation and experiment are reasonable. Although the peak concentration positions predicted by the mechanism are slightly higher than the experiments, the values are close to the experimental data. The current results suggest that the major pathways for *iso*-octane oxidation are mostly captured by the current mechanism.

3.3.4. Benzene (C₆H₆) flame

Two benzene flames from different groups were used to validate the proposed mechanism for PAH formation with aromatic fuels. Figure 11a shows comparisons with measured species profiles in benzene flames of Yang et al. [51]. Good agreement can be obtained

with the current mechanism. Since A₁ was used as the fuel, the production of C₂H₂ and C₅H₅ can solely be attributed to the A₁ decomposition, and both are very important species for larger PAH species production. The soot modeling approach adopted in the current study considers A₄ as the soot formation precursor, and steps related to A₄ consumption and soot formation are also included and considered in the CFD code for engine combustion and emission simulations. Figure 11b shows the predictions of the mechanism in another benzene flame measured by Defoeux et al. [52]. Again the trends are captured well. Note that uncertainties in the experimental measurements can be as high as one or two orders of magnitude, thus such results can still be regarded as acceptable.

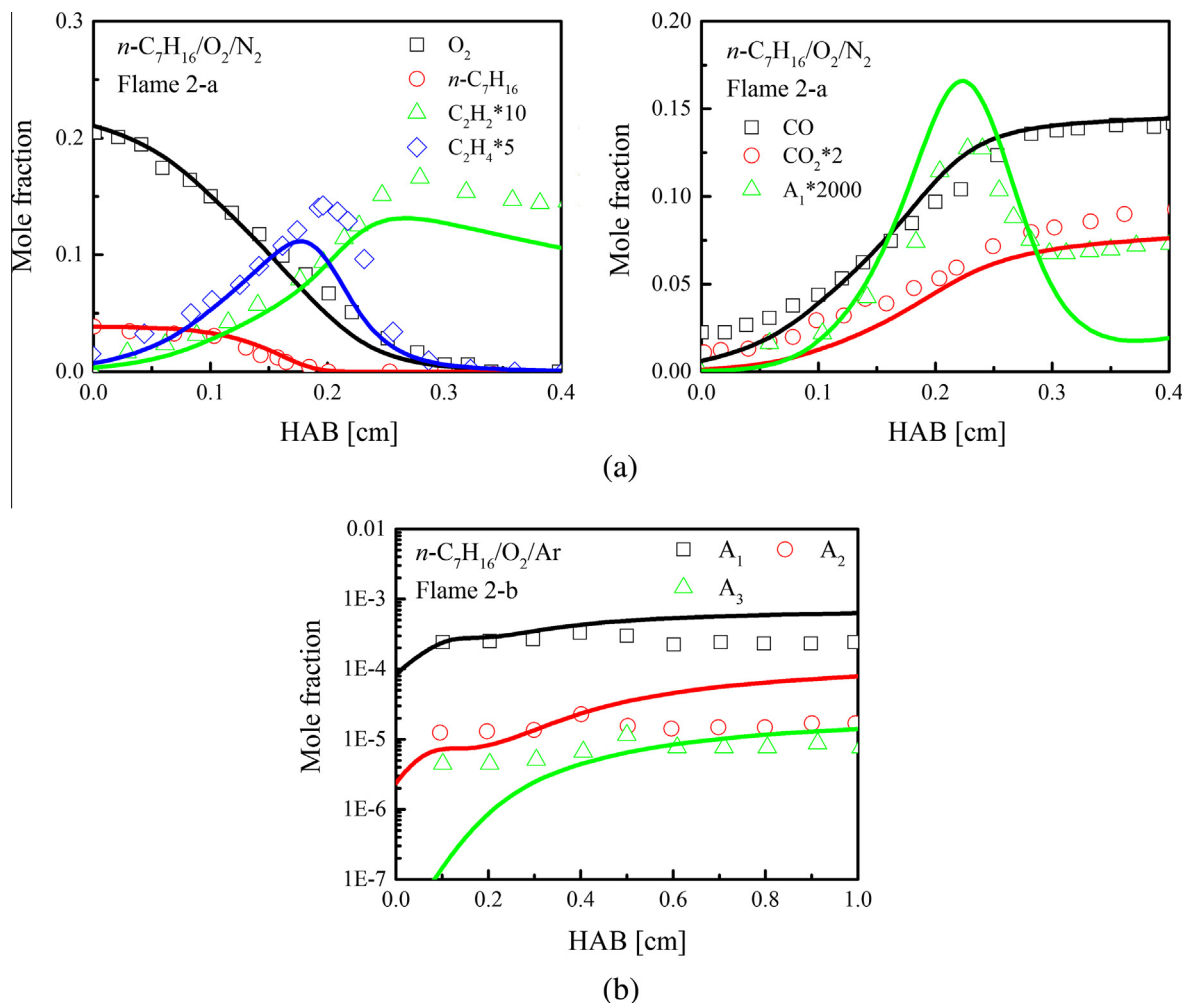


Fig. 9. Predictions of species profiles with the current mechanism in *n*-heptane flames. (a) Flame 2a and (b) Flame 2b. Experimental data are taken from [16,49,50]. (Flame 2 in Table 1).

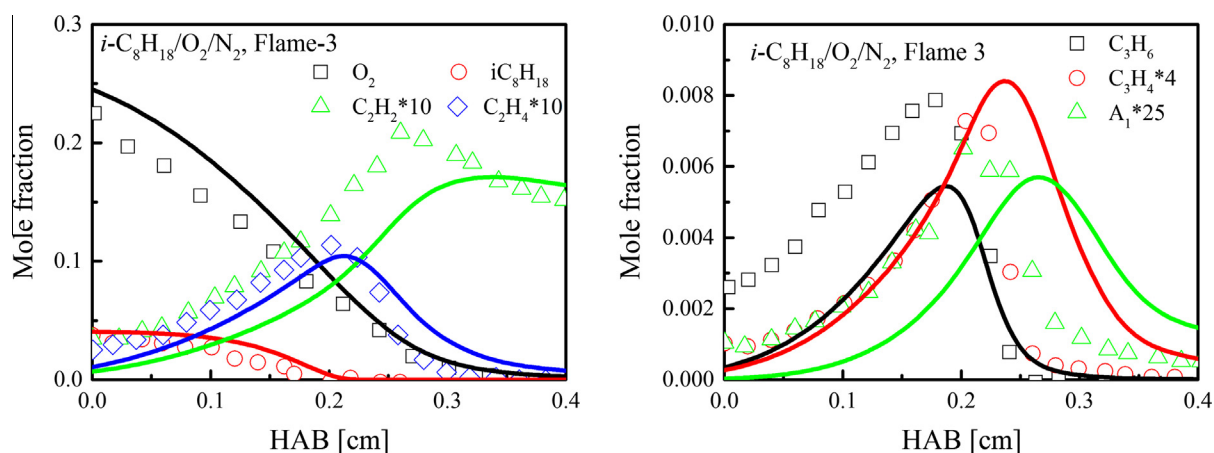


Fig. 10. Predictions of species profiles with the current mechanism in *iso*-octane flame. Experimental data are taken from [16,49] (Flame 3 in Table 1).

3.3.5. Toluene flame

Species profiles in neat toluene and CH_4 + toluene flames were also used to validate toluene oxidation and PAH formation processes with the present mechanism. Figure 12 shows the predictions in toluene flames measured by Li et al. [17] with equivalence ratio of unity. Only the important species related to PAH formation are presented here. It is seen that the mechanism tends to slightly

over predict the A_1 concentration, but the other three species agree quite well with experiments. In addition to neat toluene, species profiles from CH_4 + toluene flames studied by El Bakali et al. [53] were also compared with the current predictions, as shown in Fig. 13. It is seen that the overall agreements are still reasonable for both important intermediate species and the major oxidation products.

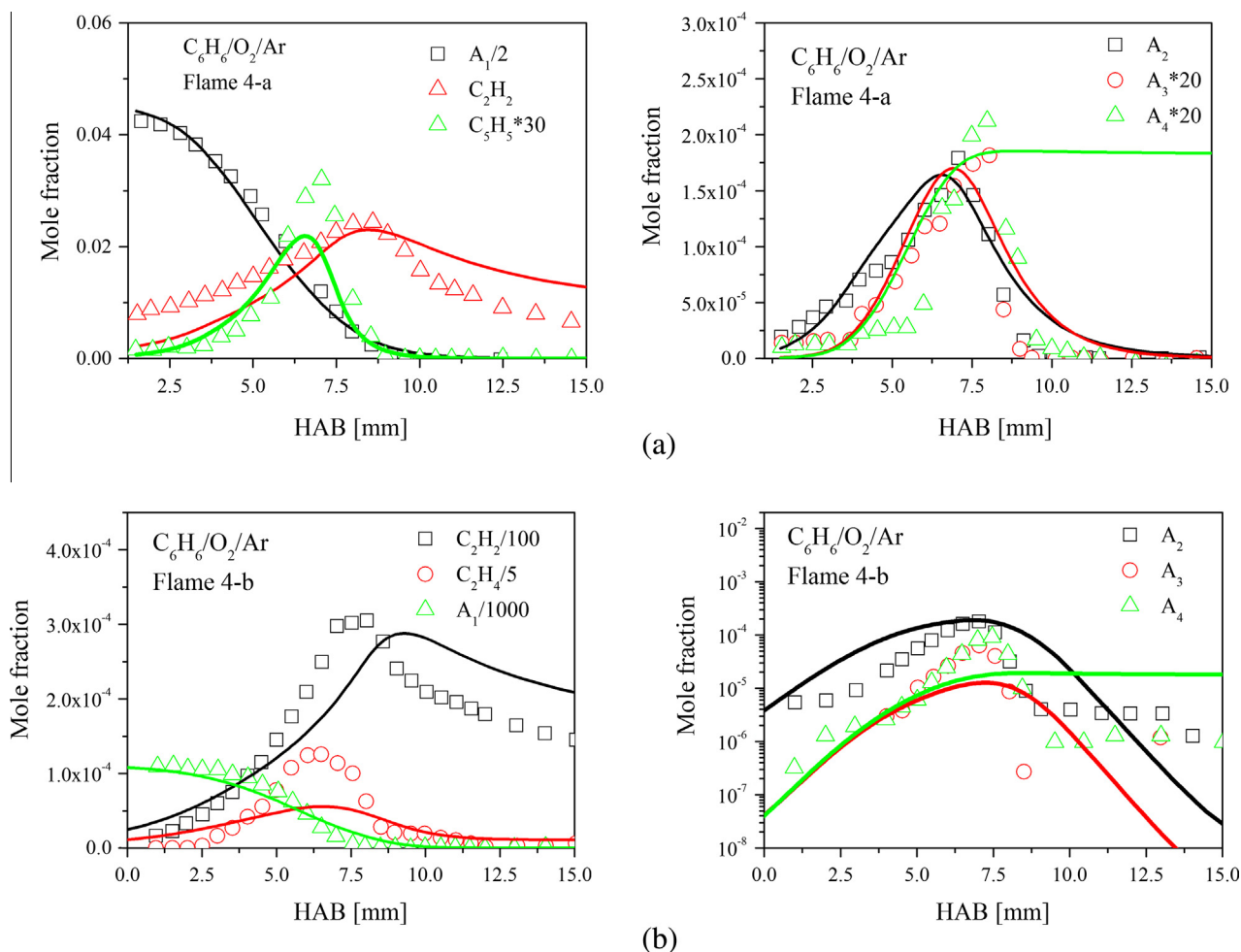


Fig. 11. Predictions of species profiles with the current mechanism in benzene flames. (a) Flame 4a and (b) Flame 4b. Experimental data are taken from [51,52] (Flame 4 in Table 1).

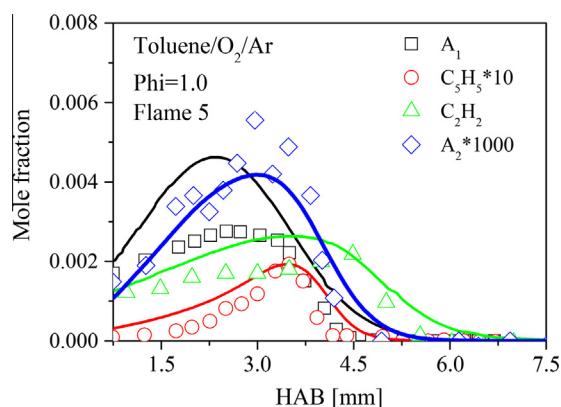


Fig. 12. Predictions of species profiles with the current mechanism in toluene flames. Experimental data are taken from [17] (Flame 5 in Table 1).

3.4. Engine HCCI combustion

3.4.1. PRF HCCI combustion

The base PRF mechanism has been validated against PRF HCCI combustion experiments conducted by Dempsey et al. [54] in the authors' previous work, thus no further results are presented here, and the related validation results can be found in [20,37]. Generally the PRF mechanism is able to predict HCCI combustion processes of PRF mixtures over a wide range of intake temperature conditions.

3.4.2. *n*-heptane/toluene HCCI combustion

The *n*-heptane/toluene engine HCCI experiments conducted by Wang et al. [30] were also taken to validate the current mechanism. Details about the engine tests can be found in [30]. Table 2 shows the engine specifications used for both HCCI and DIC engine experiments.

Table 3 shows the operating conditions for the HCCI experiments. Mixtures of 80% *n*-heptane + 20% (vol.) toluene (TRF20) and 60% *n*-heptane + 40% (vol.) toluene (TRF40) were applied in the experiments. The intake pressure and intake temperature were fixed at 0.16 MPa and 373 K, and the EGR was increased from 0% to 60% in 15% steps. The pressure traces of 50 individual cycles were captured by a pressure transducer and were averaged to get the averaged in-cylinder pressures.

The KIVA CFD code [55] was applied and coupled with the proposed mechanism for the HCCI combustion predictions. The computational mesh with around 20,000 cells for both the HCCI and DIC combustion simulations is shown in Fig. 14. Grid dependency analysis was conducted and demonstrated minimal effects of grid resolution. The sub-models applied in the simulations are listed in Table 4, and the SpeedChem chemistry solver was applied to improve computational efficiency [56]. In the HCCI simulations, a homogeneous charge was assumed in the computational domain at the start of the simulation (intake valve closing (IVC)), and the simulation ended at exhaust valve opening (EVO). Since there is no spray injection and evaporation in HCCI combustion, the spray

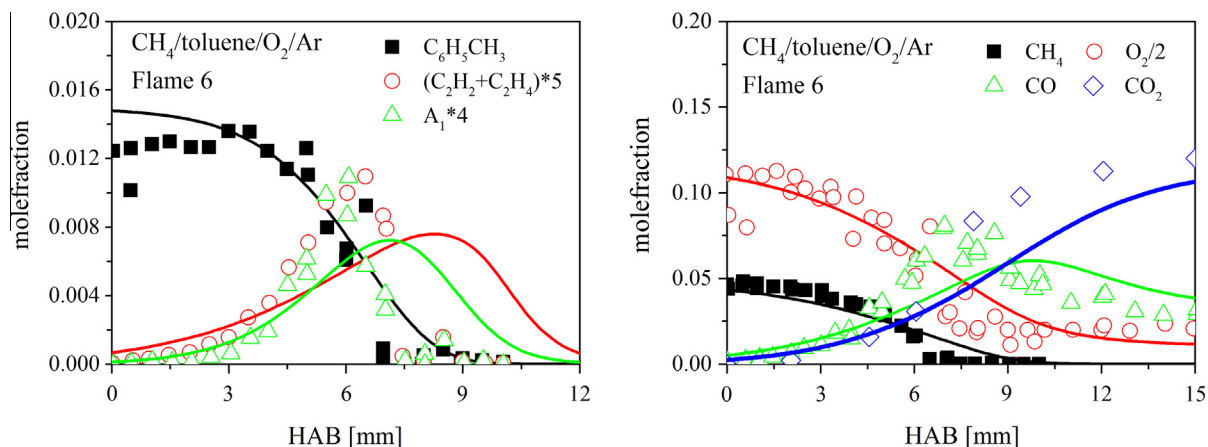


Fig. 13. Predictions of species profiles with the current mechanism in CH₄-toluene flames. Experimental data are taken from [53] (Flame 6 in Table 1).

Table 2

Engine specifications.

Engine type	Single cylinder, four stroke
Bore*stroke	105 mm × 125 mm
Displacement	1.081 L
Compression ratio	15.8:1
Combustion chamber	Bowl in piston
Fuel injection type	Common rail injection system
Injector	8 holes/injection angle 150°, 0.15 mm nozzle

Table 3

HCCI engine operating conditions.

Engine speed (r/min)	1400
IMEP (bar)	~3.5 bar
Intake air pressure (bar)	1.6
Intake temperature (K)	373
EGR ratio (%)	15, 30, 45, 60

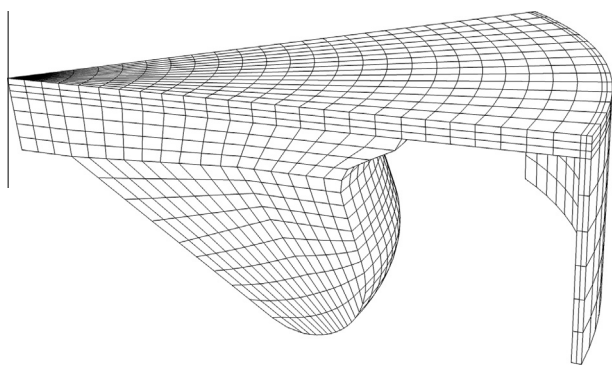


Fig. 14. Computational grid.

breakup and evaporation sub-models were not used in those simulations.

Figure 15 shows comparisons of in-cylinder pressures between the simulations and experiments at three different EGR rate conditions for the TRF20 and TRF40 fuels used in the HCCI experiments. It is seen that although the simulations tend to slightly over predict the peak in-cylinder pressures, the ignition timings and pressure rise rate are well under all tested conditions. Due to the high intake temperatures, ignition of TRF20 occurs at around −22.5 crank angle degrees after top dead center (CA ATDC). The mechanism is also able to predict the ignition and combustion processes under very low ambient oxygen concentration conditions (the initial

Table 4

Sub-models used in the HCCI and DICI simulations.

Phenomenon	Model	Reference
Spray breakup	KH-RT instability	[57,58]
Evaporation	Discrete multi-component fuel (DMC)	[59]
Turbulence	Renormalized $k-\epsilon$ model	[60]
Combustion	Sparse analytical Jacobian (SpeedChem)	[56]
Droplet collision	Radius of influence (ROI) model	[61]
Near nozzle flow	Gas-jet model	[62]

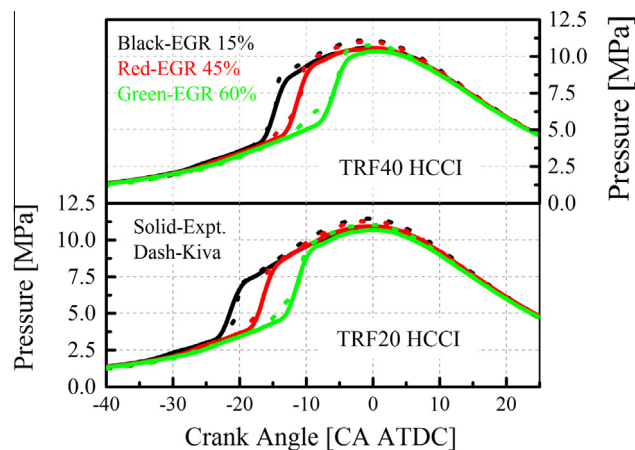


Fig. 15. Comparisons of in-cylinder pressures between experiments and simulations in TRF HCCI combustion.

oxygen concentrations for both TRF20 and TRF40 were reduced to about 10% when the EGR was increased to 60%.

4. Kinetic analysis of PAH formation pathways

4.1. Sensitivity analysis for C₂H₄

Figure 16 shows sensitivity analysis results for the four representative PAH species, A₁, A₂, A₃ and A₄ at the 2.5 mm HAB position in the C₂H₄ flame (Flame 1 in Table 1). The top ten reactions with the highest normalized sensitivity coefficients are shown for each PAH species. It is seen that for all four PAH species, it is the reactions that favor the formation of C₂H₂ that show the highest positive sensitivity, while the reactions which convert C₂H₄, C₂H₃ and

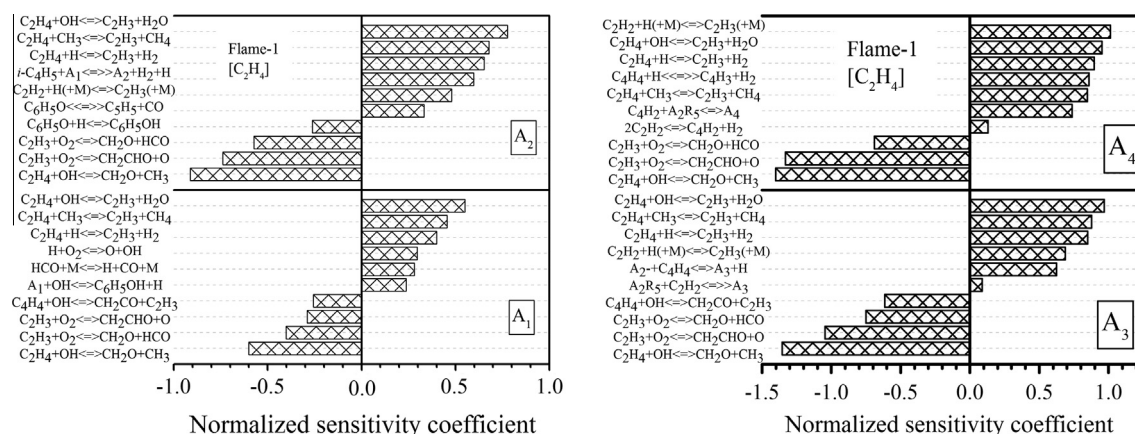


Fig. 16. Sensitivity analyses for A_1 , A_2 , A_3 and A_4 in C_2H_4 flames (Flame 1 in Table 1) at 2.5 mm HAB position.

C_2H_2 to oxygenated products, such as CH_2O and CH_2CHO , have the highest negative sensitivity. This is totally understandable since C_2H_2 is one of the most important species in the PAH building block and PAH species formation reactions. $C_2H_4 + OH = CH_2O + CH_3$ has the highest negative sensitivity while $C_2H_4 + OH = C_2H_3 + H_2O$ has the highest positive sensitivity for all four PAH species. The reactions related to the PAH building block, such as $2C_2H_2 = C_4H_2 + H_2$ and $C_4H_4 + OH = CH_2CO + C_2H_3$, also affect PAH formation. The important PAH formation pathways, including the HACA sequence and the small radical and molecule addition sequences, also show high sensitivities for the formation of larger PAH species.

4.2. Sensitivity analysis for *n*-heptane and iso-octane

Figure 17 shows sensitivity analysis results for A_1 , A_2 , A_3 and A_4 at the 2.5 mm position in the *n*-heptane flame (Flame 2b Table 1). The sensitivity in the *n*-heptane flame is similar to that of the C_2H_4 flame, where the reactions removing C_2H_4 , C_2H_3 and C_2H_2 from the system show the highest negative sensitivities, and the reactions that promote C_2H_2 production show mostly positive effects. In addition, it is seen that the core H_2 – O_2 reaction system starts to play an important role in the PAH formation process, as $O + H_2 = H + OH$ shows the highest positive sensitivity for all four PAH species. Similar trends can be seen in iso-octane flames, which are not presented here, in that the reactions involving C_2 hydrocarbon species show high sensitivity to the PAH formation.

Sensitivity analyses conducted in the C_2H_4 and *n*-heptane flames clearly indicate the importance of the reaction pathway for C_2 hydrocarbon species, especially C_2H_2 , and the core H_2 mechanism for PAH formation. These results can also be potentially used to explain the effects of oxygenated fuels on PAH and soot formation, since once the C_2 hydrocarbon species are converted to oxygenated products, they no longer contribute to PAH and soot formation [63].

4.3. Sensitivity analysis for aromatic fuels

Figure 18 shows the sensitivity analysis results for A_1 , A_2 , A_3 and A_4 at the 2.0 mm HAB position in the toluene flame (Flame 5 in Table 1). A_1 mainly comes from the ring structure of the toluene molecule. Thus, reactions with relatively high sensitivities for A_1 are those toluene consumption reactions. $C_5H_5 + C_5H_5 = A_2 + 2H$, the most important A_2 formation reaction, depends on the availability of C_5H_5 , which is determined by the competition between $C_6H_5O = C_5H_5 + CO$ and $C_6H_5CO + O = OC_6H_4O + H$. Thus, these two reactions show the highest sensitivity. A_3 and A_4 are mainly produced through the reactions between aromatic radicals and molecules. In this case, the $A_1C_2H + A_1 = A_3 + H$ and $A_2 + A_1 = A_4 + H + H_2$ reactions. The small radical and molecule addition reactions, such as $A_2 + C_4H_2 = A_3$ and $A_1 + i-C_4H_5 = A_2 + H_2 + H$, also show positive contributions to A_3 and A_4 formation. In the current approach the PAH species are formed in sequential order.

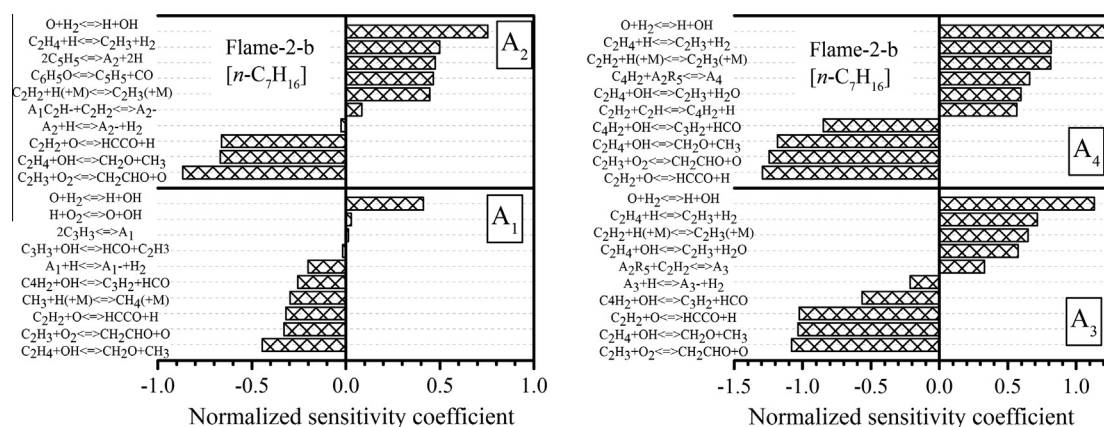


Fig. 17. Sensitivity analyses for A_1 , A_2 , A_3 and A_4 in *n*-heptane flames (Flame 2-b in Table 1) at 2.5 mm HAB position.

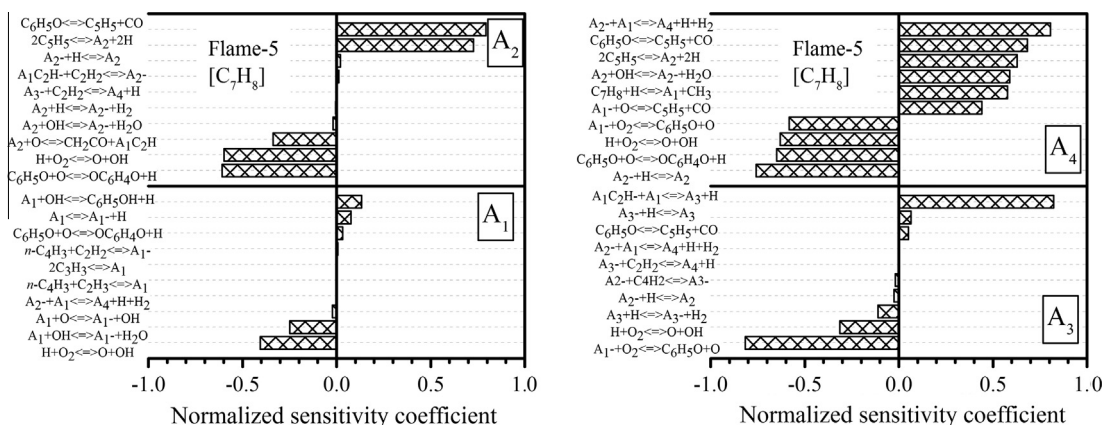


Fig. 18. Sensitivity analyses for A_1 , A_2 , A_3 and A_4 in toluene flames (Flame 5 in Table 1) at 2.0 mm HAB position.

Therefore the formation of larger PAH molecules highly depends on the low level PAH species formation. $A_1 + O_2 = C_6H_5O + O$ shows high negative sensitivity for A_3 due to the fact that the formation of A_3 depends on A_1 , while this reaction actually removes A_1 from the system. The PAH sensitivity results for benzene are very similar to those of toluene, and thus are not presented here.

Table 5 shows the important reactions related to PAH formation for C_2H_4 , n -heptane, iso -octane, benzene and toluene. The reactions are listed in descending order of importance in this table for each PAH species for all the fuels.

For non-aromatic fuels A_1 is mainly produced through the combination of $C_4 + C_2$ and two C_3 radicals, while for aromatic fuels, A_1 either directly comes from the fuel, or comes from the oxidation products of the fuel (toluene). The major formation reactions for A_2 are quite similar for all the fuels, including the addition reactions between radicals and molecules, $i-C_4H_5 + A_1 = A_2 + 2H + H$ and $A_1 + C_4H_3 = A_2$, and $C_5H_5 + C_5H_5 = A_2 + 2H$. The difference is that for non-aromatic fuels, C_5H_5 is mainly formed through the combination of C_2 and C_3 species, while for aromatic fuels, $C_6H_5O = C_5H_5 + CO$ is the most important reaction that produces C_5H_5 .

A_3 is mainly produced through the addition reactions between small radicals and molecules and the HACA scheme for non-aromatic fuels, including $A_2 + C_4H_4 = A_3 + H$ and $A_2R_5 + C_2H_2 = A_3$, while for aromatic fuels, the reactions between aromatic radicals and molecules are more important. The important formation pathways for A_4 are quite similar for both non-aromatic and aromatic fuels, where the addition reactions between radicals and molecules, reactions between aromatic radicals and molecules, and also the HACA mechanism, all play roles in the A_4 formation process.

Table 5
Major PAH formation pathways for tested fuels.

Fuel	C_2H_4	$n-C_7H_{16}$	$i-C_8H_{18}$	C_6H_6	$C_6H_5CH_3$
A_1	$C_2H_3 + C_4H_4 = A_1 + H$ $i-C_4H_5 + C_2H_2 = A_1 + H$ $C_4H_3 + C_2H_3 = A_1$	$i-C_4H_5 + C_2H_2 = A_1 + H$ $C_2H_3 + C_4H_4 = A_1 + H$ $2C_3H_3 = A_1$	$2C_3H_3 = A_1$ $i-C_4H_5 + C_2H_2 = A_1 + H$ $C_2H_3 + C_4H_4 = A_1 + H$	–	$C_7H_8 + H = A_1 + CH_3$ $A_1 + OH = C_6H_5OH + H$
A_2	$i-C_4H_5 + A_1 = A_2 + 2H + H$ $2C_5H_5 = A_2 + 2H$	$2C_5H_5 = A_2 + 2H$ $i-C_4H_5 + A_1 = A_2 + 2H + H$	$2C_5H_5 = A_2 + 2H$ $i-C_4H_5 + A_1 = A_2 + 2H + H$	$i-C_4H_5 + A_1 = A_2 + 2H + H$ $2C_5H_5 = A_2 + 2H$ $A_1 + C_4H_3 = A_2$	$2C_5H_5 = A_2 + 2H$ $i-C_4H_5 + A_1 = A_2 + 2H + H$ $A_1 + C_4H_3 = A_2$
A_3	$A_2 + C_4H_4 = A_3 + H$ $A_2R_5 + C_2H_2 = A_3$	$A_2 + C_4H_4 = A_3 + H$ $A_2R_5 + C_2H_2 = A_3$	$A_2 + C_4H_4 = A_3 + H$ $A_2R_5 + C_2H_2 = A_3$	$A_1C_2H + A_1 = A_3 + H$ $A_1C_2H + A_1 = A_3 + H$	$A_1C_2H + A_1 = A_3 + H$ $A_1C_2H + A_1 = A_3 + H$
A_4	$C_4H_2 + A_2R_5 = A_4$ $A_2 + A_1 = A_4 + H + H_2$ $A_3 + C_2H_2 = A_4$	$C_4H_2 + A_2R_5 = A_4$ $A_2 + A_1 = A_4 + H + H_2$	$C_4H_2 + A_2R_5 = A_4$ $A_2 + A_1 = A_4 + H + H_2$	$A_2 + A_1 = A_4 + H + H_2$ $A_1C_2H + A_1C_2H = A_4 + H$ $C_4H_2 + A_2R_5 = A_4$	$A_2 + A_1 = A_4 + H + H_2$ $A_1C_2H + A_1C_2H = A_4 + H$ $A_2 + A_1 = A_4 + H_2$

5. Application of the proposed mechanism in a DICI engine for soot prediction

5.1. Experimental and simulation methods

The present TRF mechanism was also applied for combustion and soot emission predictions in a n -heptane/toluene fueled DICI engine. The engine is the same engine that was used to run the n -heptane/toluene HCCI experiments. The engine specifications and test-bed setup have already been discussed in the HCCI combustion simulation section. The DICI engine operating conditions are listed in Table 6.

Diesel, n -heptane and mixtures of n -heptane and toluene (TRF15: 85% n -heptane and 15% (vol.) toluene and TRF30: 70% n -heptane and 30% (vol.) toluene) were tested in the experiments. During the experiments CA10 (crank angle at 10% accumulated

Table 6
DICI Engine operating conditions.

Engine speed (r/min)	1400
IMEP (bar)	~8.3 bar
Intake air pressure (bar)	1.8
Intake temperature (K)	313
EGR ratio (%)	0–50%
Fuel	Diesel, n -heptane, TRF15, TRF30
SOI	–5.3 to –9.2 deg ATDC
CA10	Top dead center (TDC)
Injection pressure (bar)	1000
Fuel mass	50 mg diesel energy

heat release) was fixed at top dead center (TDC) by adjusting the start-of-injection (SOI) timing when the EGR was increased from 0% to 50%. The two TRF fuels were chosen to highlight the effects of toluene content on the mixing, combustion and soot formation processes. NO_x was measured with an AVL FTIR gas emission analyzer and soot was measured with AVL 415S smoke meter and was then converted to specific soot emissions (g/kW h) through the equation provided in the specification of the smoke meter [64].

The computational mesh and sub-models applied in the simulations were shown in Fig. 14 and Table 4, and the simulations started at IVC and ended at EVO. The GasJet model and ROI collision model were applied in the simulations to reduce numerical grid dependency [61,62]. The physical properties of diesel fuel were represented with *n*-tetradecane (C₁₄H₃₀) and the *n*-heptane mechanism was applied to model the combustion process for diesel, while the other three fuels' physical and chemical properties were represented with their own properties. A multi-step soot model, which considers A₄ as the soot precursor, was applied for soot prediction. This soot modeling approach considers soot inception through A₄, soot surface growth assisted by C₂H₂ and PAH species (A₁, A₂, A₃ and A₄), soot oxidation through oxygen and OH radicals and soot particle coagulation. Good agreements in soot volume fraction and the location of the soot formation region have been achieved by applying this soot modeling approach in Sandia *n*-heptane spray combustion simulations. Details of the soot model can be found in [34,65].

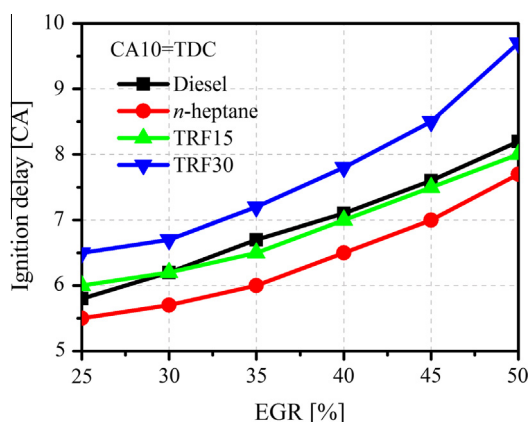


Fig. 19. Experimental ignition delays as function of EGR rate for the tested fuels in DI CI combustion.

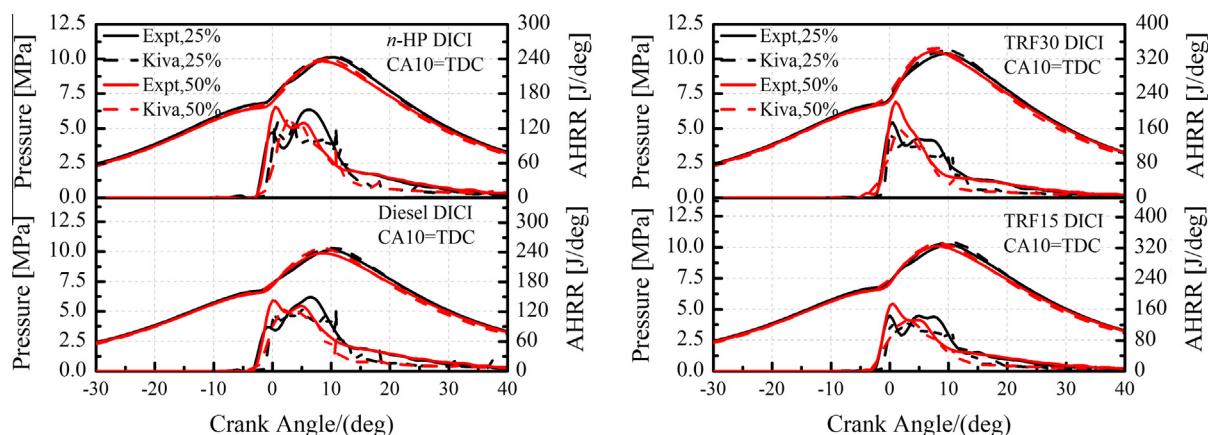


Fig. 20. Comparisons of in-cylinder pressures and AHRRs between simulation and experiment for the tested fuels in DI CI combustion.

5.2. In-cylinder pressure and apparent heat release rate (AHRR)

Since the combustion characteristics and emissions of the tested fuels were found to be similar when the EGR was below 25%, only the results with EGR higher than 25% are presented here. Figure 19 shows the ignition delays of the various fuels from the experiments. The ignition delay is defined as the interval between the SOI timing (the commanded injection delay is considered) and the CA10 timing. It is clearly seen that under the same EGR condition, *n*-heptane needs the latest and TRF30 needs the most advanced SOI timing to maintain the CA10 timing at TDC, and the ignition delays of diesel and TRF15 are very similar. The ignition delays of all fuels also increase with increased EGR rate due to the reduced ambient oxygen concentration and lower compression temperature.

Figure 20 shows comparisons of the in-cylinder pressures and AHRRs between the simulations and experiments for all the tested fuels at EGR rates of 25% and 50%. It is seen that the overall agreements between the simulations and experiments are reasonably good. The premixed combustion heat release of the various fuels increases as the EGR increases due to the much advanced SOI timing and longer ignition delay, and the earliest SOI timing required by TRF30 fuel results in the most significant premixed heat release amount. The mechanism is able to accurately capture the essential combustion characteristics, even when the EGR is increased to 50%, which corresponds to an ambient oxygen concentration of about 12.8%.

5.3. NO_x and soot emissions

Figure 21 shows comparisons of NO_x and soot emissions between the simulations and experiments, and the corresponding ambient oxygen concentrations for each EGR rate are also listed in the figure. Though there are discrepancies in the absolute values, the trends are well captured by the simulations for both NO_x and soot emissions. The general trend for NO_x emission is that under the same EGR rate condition, fuels with advanced SOI timing and longer ignition delays (lower cetane number) produce more NO_x, as shown in both the experiments and simulations. The simulations also capture the soot emission trends. The addition of toluene to *n*-heptane is helpful to reduce the cetane number and to prolong the ignition delay. This improves the mixing process, which is beneficial for soot reduction. However, the introduction of toluene also greatly enhances the PAH and soot formation due to its ring structure. Both the experiments and simulations show that diesel fuel produces the highest soot, followed by TRF15 and TRF30, and *n*-heptane produces the lowest soot emission. It is

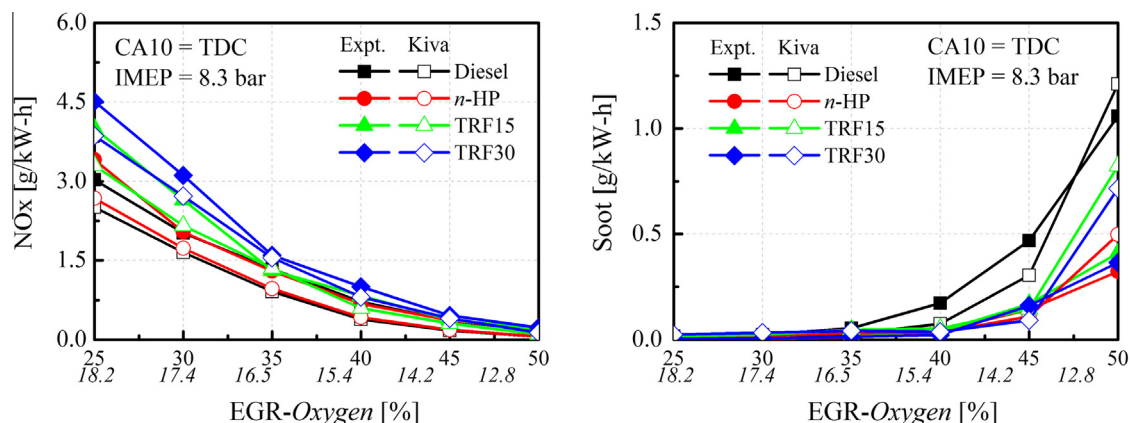


Fig. 21. Comparisons of NOx and soot emissions between experiments and simulations in DICI combustion.

worth pointing out that, different from the authors' previous results [30], in the current study the same soot model constants were applied for all the tested fuels, which is a considerable improvement over the previous mechanism which required different constants for each fuel. This indicates that the important reaction pathways related to the fuel decomposition and oxidation and

also the PAH formation for the tested fuels are considered and are included in the current mechanism.

In the current simulations the combustion processes of both diesel and *n*-heptane are represented with the *n*-heptane mechanism, and differences in the results are due to the fuel's physical properties and SOI timings. Since the volatility of *n*-heptane is

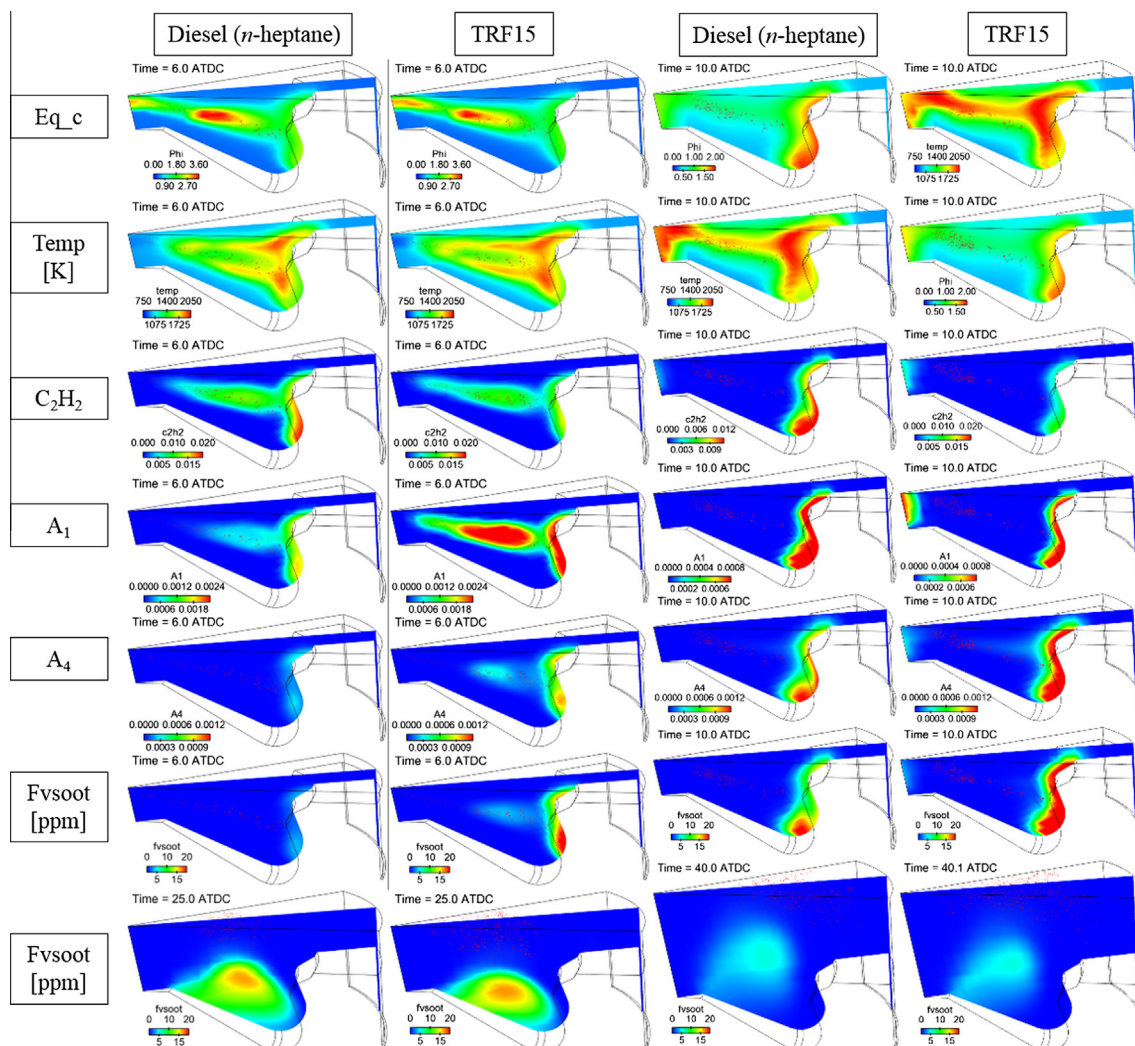


Fig. 22. In-cylinder distributions of equivalence ratio, temperature and the concentrations of C_2H_2 , A_1 and A_4 , and soot volume fraction (fvsoot in ppm) for diesel and TRF15 at the 50% EGR condition.

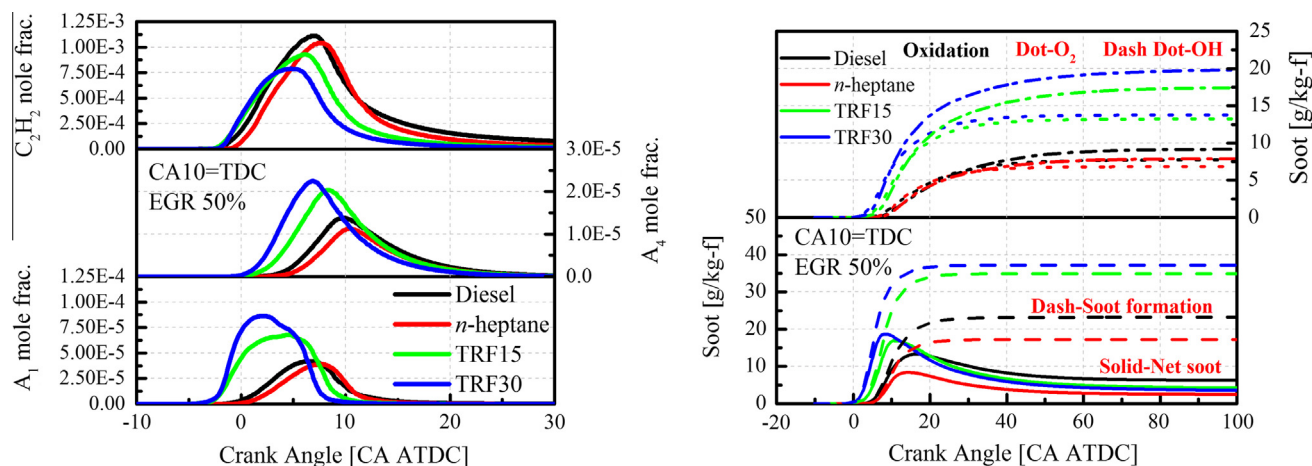


Fig. 23. Evolution processes of C_2H_2 , A_1 , A_4 and soot (net soot, soot formation and soot oxidation by O_2 and OH) in DICI combustion simulations for the tested fuels at the 50% EGR condition.

higher than *n*-tetradecane, and the ignition of *n*-heptane at the 50% EGR condition is 0.5° later than diesel (Fig. 19), the improvement of the mixing process considerably reduces the extent of the fuel rich regions and then soot formation. Similar explanations can also be applied to comparisons between diesel and TRF15 (shown below in Fig. 22). Both fuels have similar ignition delays over the tested EGR range, but the final soot emission of TRF15 is still lower than diesel.

It is also of interest to see that although the toluene content in TRF30 is higher than TRF15, the soot emissions of TRF30 from both experiment and simulation are lower than TRF15. As mentioned above, the improved mixing process of TRF30 (i.e. enhanced air entrainment) due to the lower cetane number is helpful for soot oxidation and reduction. However, the much higher toluene content in TRF30 also promotes PAH and soot formation. Thus, the final results depend on the competition between these two processes. The results suggest that under the current conditions, the improvement in mixing from TRF15 to TRF30 overcomes the adverse effects of the higher toluene content on PAH and soot formation, although the difference is quite small.

In-cylinder distributions of equivalence ratio, temperature and the concentrations of C_2H_2 , A_1 and A_4 (mass fraction) and soot volume fractions (fvsoot in ppm) for diesel and TRF15 at the 50% EGR condition are shown in Fig. 22 on the spray axis for one spray plume.

The equivalence ratio distributions clearly suggest that the mixing process of TRF15 is better than diesel, regardless their quite similar ignition delays. The C_2H_2 concentration in the TRF15 case is lower than diesel. Since A_1 is the direct decomposition product of toluene, a high A_1 region exists even in the injected spray, while A_1 in the diesel case is mostly located in the spray downstream and in near wall regions. A_4 and fvsoot with TRF15 are also higher than diesel in the early combustion stages. Due to the improved mixing, the soot oxidation for TRF15 is considerably faster than diesel. Thus, as the combustion proceeds, it is seen that the soot volume fraction of TRF15 is lower than diesel later in the expansion stroke at 40° CA ATDC, as shown in Fig. 22.

Figure 23 shows the evolution of important soot related species (C_2H_2 , A_1 and A_4) and soot at the 50% EGR condition. The evolution of net soot, soot formation and soot oxidation by oxygen and OH radicals is also shown. It is seen that the presence of toluene greatly promotes PAH formation. Both A_1 and A_4 are much higher, while C_2H_2 is lower than diesel and *n*-heptane, and the onset of PAH species is also earlier than for the non-aromatic fuels, which again implies that the ring structure in toluene is an important

source of A_1 and PAH formation. There are two possible reasons for the lower C_2H_2 concentrations in the TRF fuels compared to diesel and *n*-heptane. One is the improved mixing for the TRF fuels due to their longer ignition delays and the other is the different C_2H_2 formation pathways between *n*-heptane and toluene. C_2H_2 formation in toluene is difficult due to the very stable ring structure, which mainly relies on the ring opening reactions, $C_6H_5(A_1) \rightarrow C_6H_5O \rightarrow OC_6H_4O \rightarrow C_2H_2$, as can be seen in Fig. 3. It is also seen that soot formation with the TRF fuels is much faster and higher than with non-aromatic fuels. However, the soot oxidation rates through both oxygen and OH radicals in the TRF fuels are also higher than for diesel and *n*-heptane in the late combustion phase due to the enhanced mixing and longer flame lift-off lengths of the TRF fuels. Since the net soot is determined by the competition of soot formation and oxidation, the enhanced soot oxidation for the TRF fuels plays more important roles and results in lower soot compared to diesel.

6. Conclusions

In the current study a reduced TRF-PAH mechanism with 109 species and 543 reactions is proposed for combustion, PAH and soot formation predictions. Experimental data from shock tube ignition delays, laminar flame speeds, species profiles in eight flames and in engine HCCI combustion have been used to validate the proposed mechanism. Sensitivity and reaction pathway analyses were conducted to identify the important reactions and reaction pathways for PAH formation with both non-aromatic and aromatic fuels. The mechanism was also applied to *n*-heptane/toluene DICI engine combustion and soot emission predictions. The major conclusions can be summarized as follows:

1. A reduced *n*-heptane/*iso*-octane/toluene/PAH mechanism with 109 species and 543 reactions is presented for combustion, PAH and soot formation predictions. Ignition delays, laminar flame speeds, species profiles from eight premixed flames and engine HCCI combustion data were used to validate the mechanism and the agreements between predictions and simulations are mostly reasonable. The good agreement in predictions of PAH and soot-related species with the various fuels suggests that the current mechanism can be used for PAH predictions for TRF fuels in practical applications.
2. For non-aromatic fuels, reactions related to C_2H_4 , C_2H_3 and C_2H_2 , such as $C_2H_4 + OH = CH_2O + CH_3$ and $C_2H_4 + OH = C_2H_3 + H_2O$, show the highest sensitivity for PAH formation while for

- aromatic fuels, PAH formation depends highly on reactions related to the ring structure, such as $C_6H_5O = C_5H_5 + CO$ and $C_6H_5O + O = OC_6H_4O + H$.
- For non-aromatic fuels, A_1 is mainly produced through the combination of $C_4 + C_2$ and two C_3 radicals, while for aromatic fuels, A_1 either directly comes from the fuel or comes from the oxidation products of the fuel (toluene). The formation pathways of higher PAH species, including A_2 , A_3 and A_4 , are similar for all the fuels considered, including $C_5H_5 + C_5H_5 = A_2 + 2H$, the HACA scheme, the addition reactions between radicals and molecules and the reactions between aromatic radicals and molecules.
 - The proposed mechanism has also been applied to model DIC spray combustion and emissions of diesel, *n*-heptane and mixtures of *n*-heptane/toluene fuels. The simulations are able to capture both the combustion characteristics and emissions trends for all tested fuels. The toluene content greatly enhances PAH and soot formation in the early combustion stages. However, the improved mixing of TRF fuels due to their longer ignition delay also favors soot oxidation, thus the final soot emissions of both considered TRF fuels (TRF15 and TRF30) were lower than diesel, which again highlights the importance of fuel properties on engine combustion and emissions.

Acknowledgments

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.combustflame.2015.02.005>.

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